LSTM and GRU comparison results report

Multi-pollutant forecasting | 4 November 2026

This report contains the saved errors and graphs for LSTM and GRU comparison.

Dataset: supplied combined hourly CSV with 11 pollutants and 14,592 rows.

Training and validation use 2025. Testing uses January to August 2026.

Daily: hourly forecasts averaged to a day versus a direct daily forecast.

Weekly: daily forecasts averaged to a week versus a direct weekly forecast.

Monthly: weekly blocks averaged to a month versus a direct monthly forecast.

MAE is the average absolute error. RMSE gives larger errors more weight.

Lower MAE and RMSE are better. Compare values within the same pollutant.

Units: CO is mg/m3; NOx is ppb; other pollutants use source micrograms/m3.

Each actual average needs 75% hourly coverage. Missing actuals are not scored.

Forecasts use measurements available before the period starts.

Later 2026 forecasts can use earlier observed 2026 data, without retraining.

Monthly findings are exploratory: only nine direct training windows and eight test months.

One seeded run does not establish superiority. AQI classification is still pending.

Network: one recurrent layer with 32 hidden units; Adam at 0.001; seed 42.

Validate on October-December 2025; select epochs, then refit on eligible 2025 data.

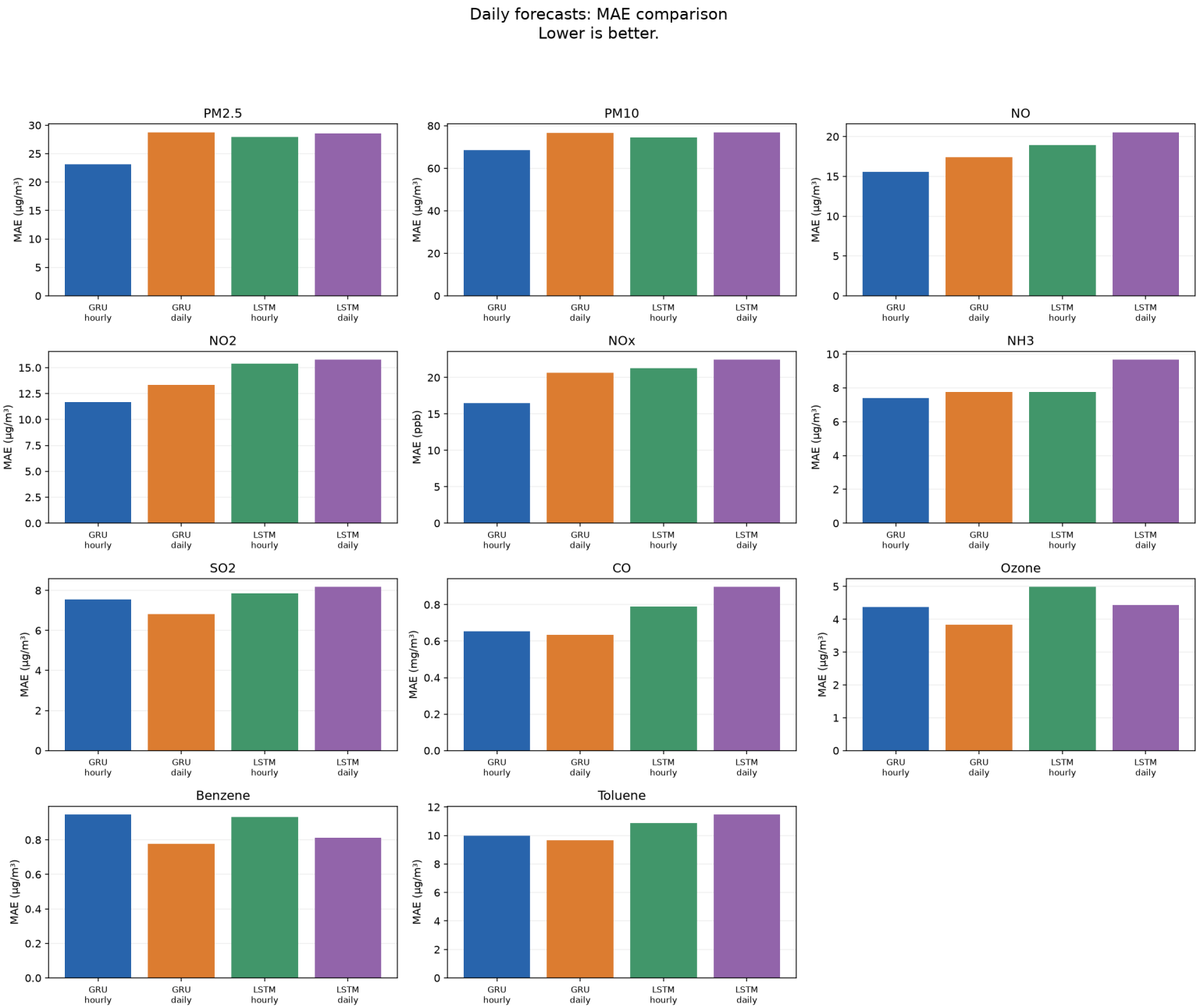
Missing past inputs fill forward or use training medians; scaling uses training data only.

LSTM and GRU comparison daily errors

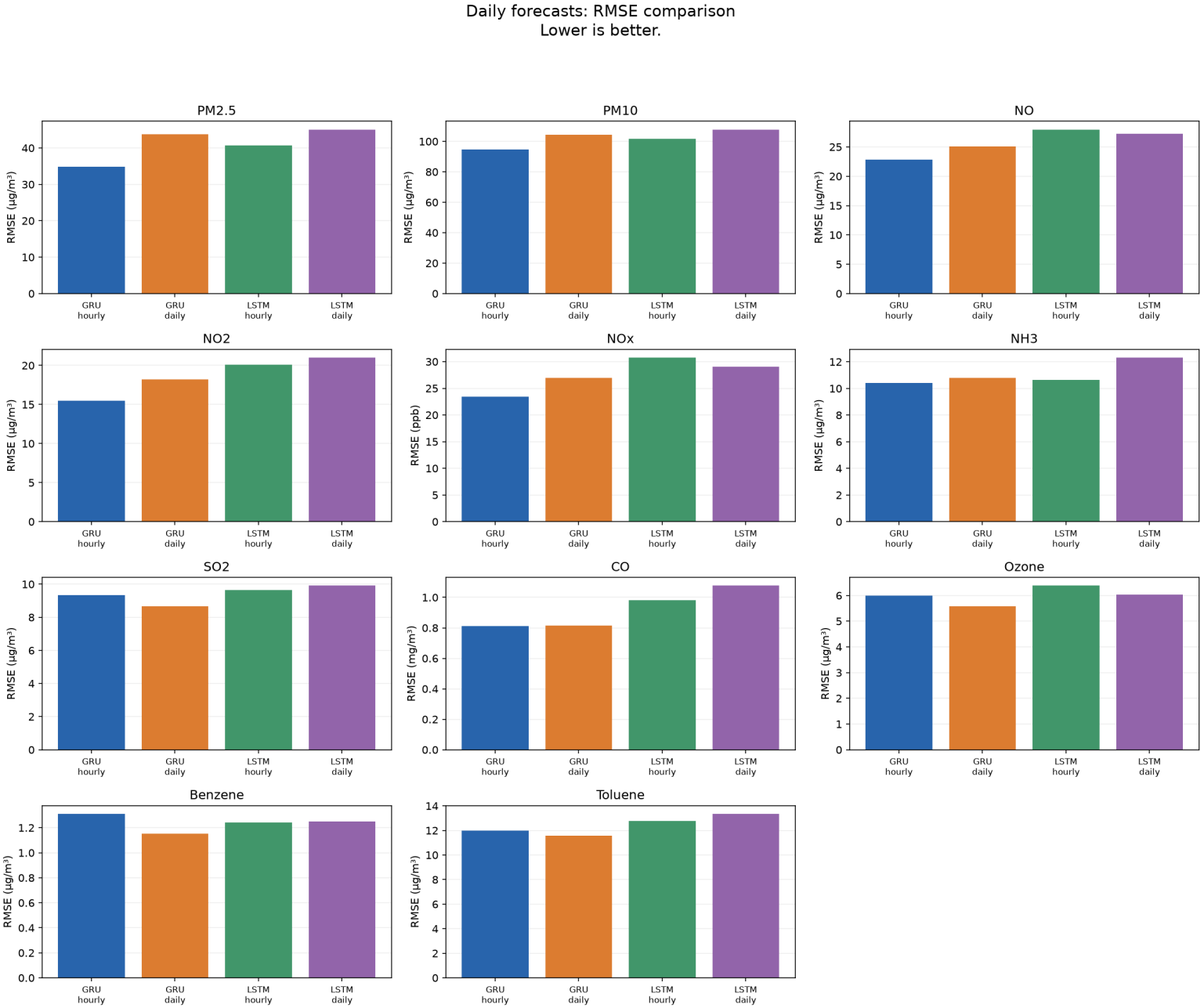
Original concentration units | N is the number of scored periods | Lower is better

Pollutant	Model	Input scale	N	MAE	RMSE
Benzene	GRU	daily	229	0.78	1.15
Benzene	GRU	hourly	229	0.95	1.31
Benzene	LSTM	daily	229	0.81	1.25
Benzene	LSTM	hourly	229	0.93	1.24
CO	GRU	daily	228	0.63	0.81
CO	GRU	hourly	228	0.65	0.81
CO	LSTM	daily	228	0.90	1.08
CO	LSTM	hourly	228	0.79	0.98
NH3	GRU	daily	229	7.76	10.80
NH3	GRU	hourly	229	7.40	10.40
NH3	LSTM	daily	229	9.68	12.32
NH3	LSTM	hourly	229	7.76	10.65
NO	GRU	daily	230	17.43	25.10
NO	GRU	hourly	230	15.57	22.80
NO	LSTM	daily	230	20.55	27.25
NO	LSTM	hourly	230	18.93	27.97
NO2	GRU	daily	227	13.33	18.24
NO2	GRU	hourly	227	11.70	15.45
NO2	LSTM	daily	227	15.78	21.01
NO2	LSTM	hourly	227	15.39	20.07
NOx	GRU	daily	231	20.66	26.98
NOx	GRU	hourly	231	16.47	23.40
NOx	LSTM	daily	231	22.46	29.09
NOx	LSTM	hourly	231	21.27	30.76
Ozone	GRU	daily	227	3.84	5.57
Ozone	GRU	hourly	227	4.37	5.99
Ozone	LSTM	daily	227	4.44	6.03
Ozone	LSTM	hourly	227	4.98	6.38
PM10	GRU	daily	227	76.71	104.22
PM10	GRU	hourly	227	68.51	94.57
PM10	LSTM	daily	227	76.97	107.84
PM10	LSTM	hourly	227	74.46	101.61
PM2.5	GRU	daily	228	28.77	43.71
PM2.5	GRU	hourly	228	23.14	34.88
PM2.5	LSTM	daily	228	28.58	45.09
PM2.5	LSTM	hourly	228	27.94	40.67
SO2	GRU	daily	226	6.81	8.65
SO2	GRU	hourly	226	7.55	9.34
SO2	LSTM	daily	226	8.17	9.91
SO2	LSTM	hourly	226	7.86	9.63
Toluene	GRU	daily	229	9.67	11.59
Toluene	GRU	hourly	229	9.98	12.00
Toluene	LSTM	daily	229	11.49	13.36
Toluene	LSTM	hourly	229	10.86	12.77

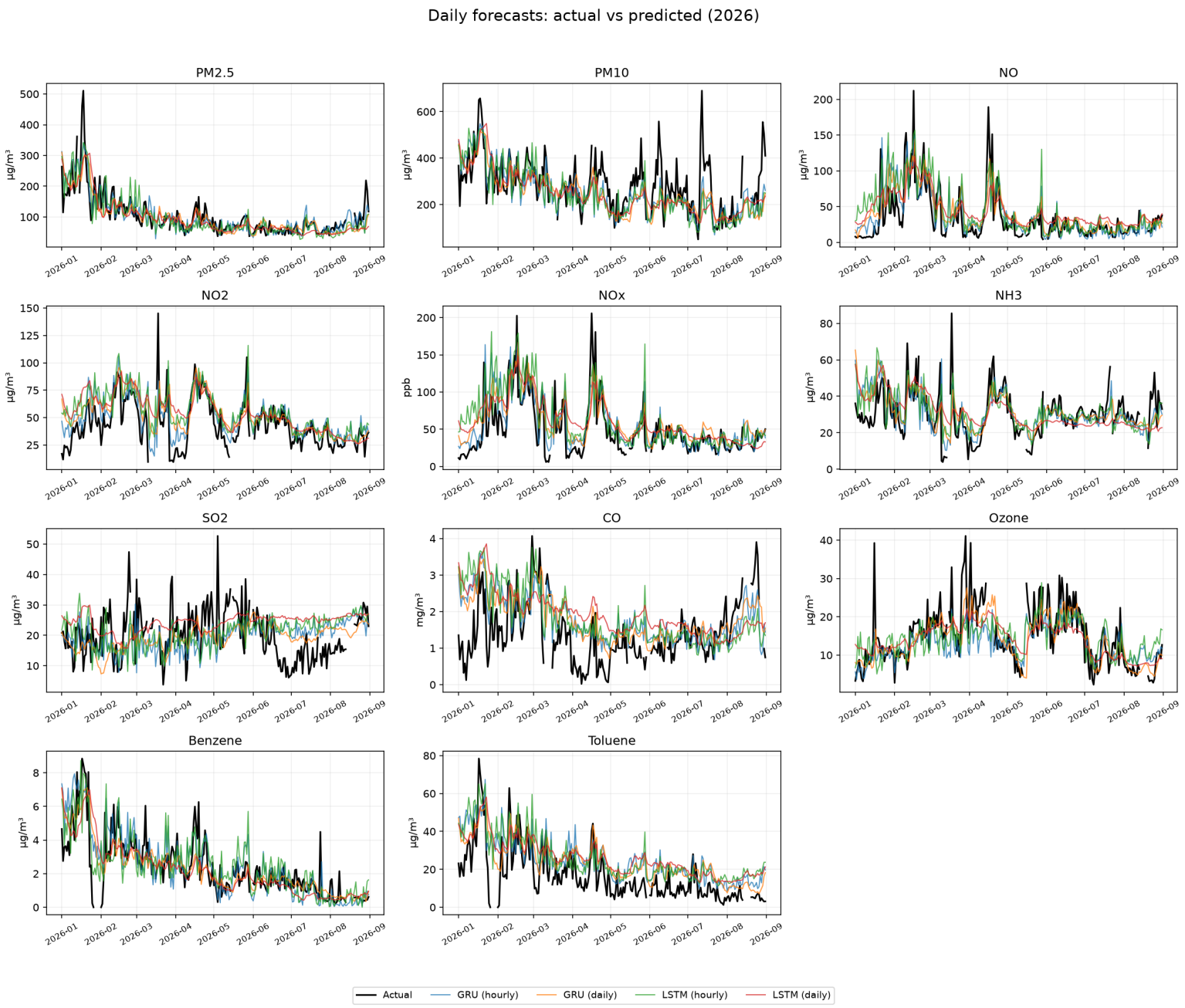
Daily MAE comparison



Daily RMSE comparison



Daily actual and predicted values

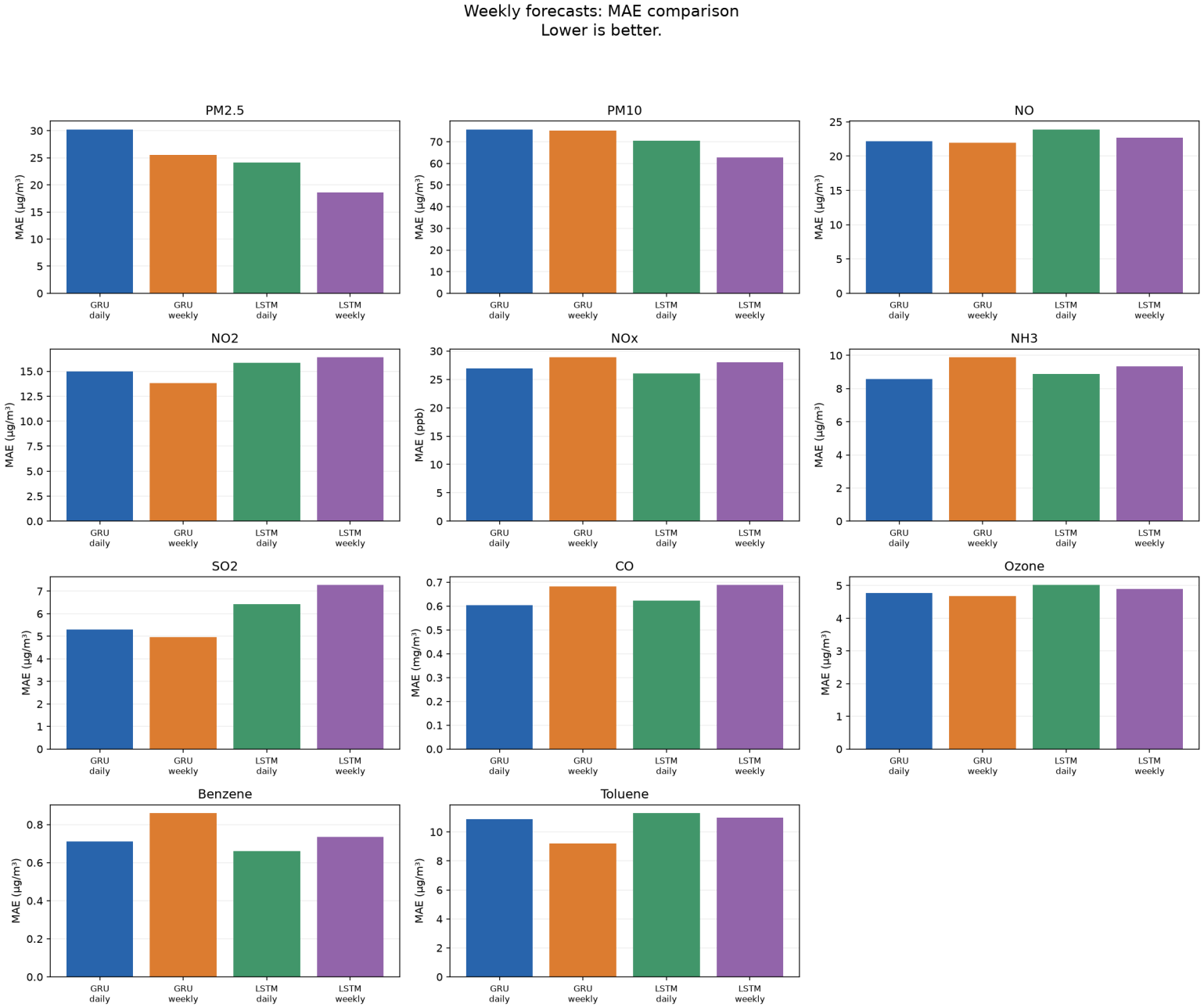


LSTM and GRU comparison weekly errors

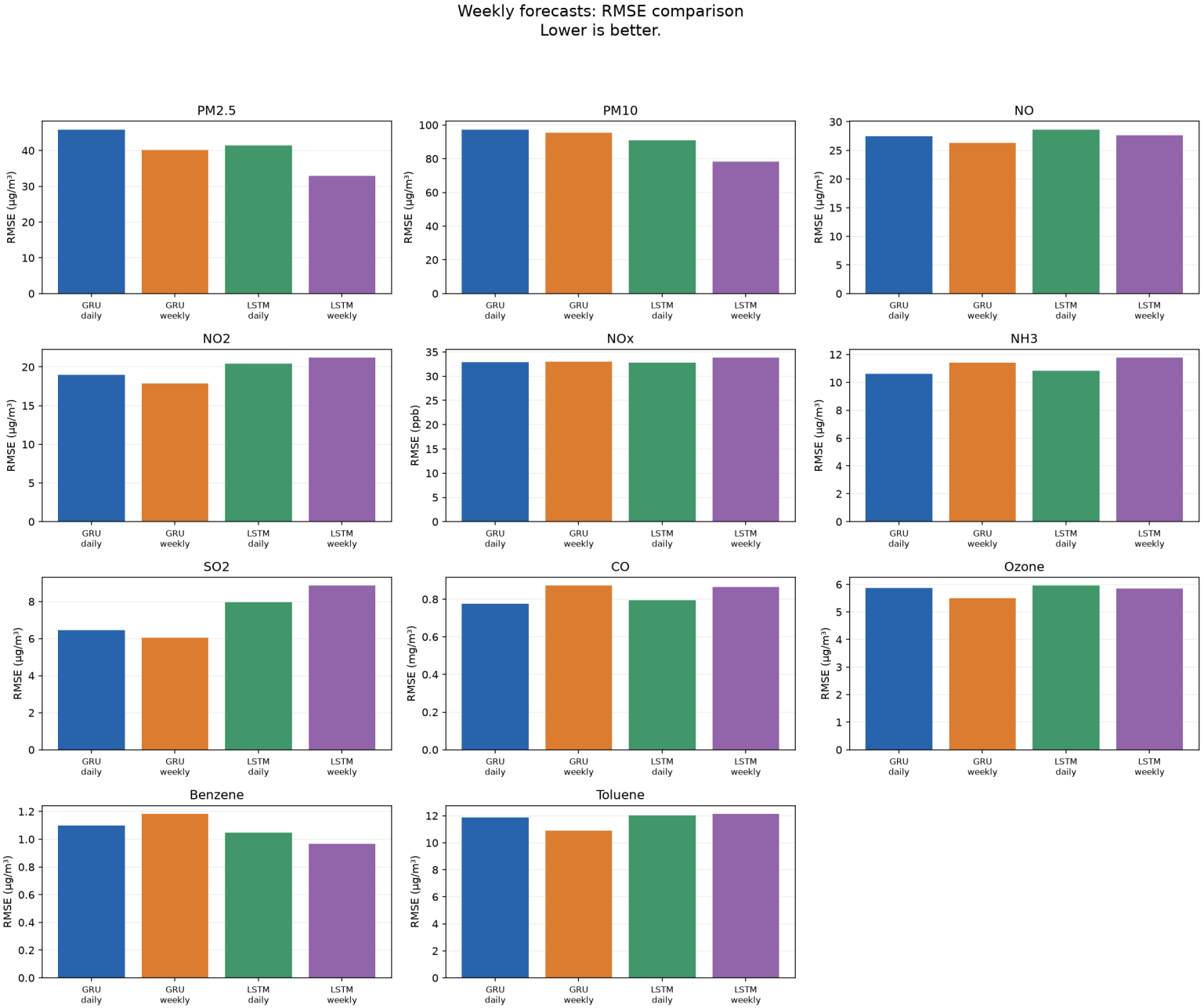
Original concentration units | N is the number of scored periods | Lower is better

Pollutant	Model	Input scale	N	MAE	RMSE
Benzene	GRU	daily	32	0.71	1.10
Benzene	GRU	weekly	32	0.86	1.18
Benzene	LSTM	daily	32	0.66	1.05
Benzene	LSTM	weekly	32	0.74	0.97
CO	GRU	daily	32	0.60	0.78
CO	GRU	weekly	32	0.68	0.87
CO	LSTM	daily	32	0.62	0.80
CO	LSTM	weekly	32	0.69	0.87
NH3	GRU	daily	33	8.59	10.62
NH3	GRU	weekly	33	9.89	11.41
NH3	LSTM	daily	33	8.89	10.83
NH3	LSTM	weekly	33	9.34	11.78
NO	GRU	daily	33	22.19	27.47
NO	GRU	weekly	33	21.98	26.35
NO	LSTM	daily	33	23.91	28.66
NO	LSTM	weekly	33	22.67	27.64
NO2	GRU	daily	32	14.96	18.98
NO2	GRU	weekly	32	13.84	17.89
NO2	LSTM	daily	32	15.85	20.44
NO2	LSTM	weekly	32	16.40	21.21
NOx	GRU	daily	33	26.97	32.89
NOx	GRU	weekly	33	28.92	32.99
NOx	LSTM	daily	33	26.08	32.84
NOx	LSTM	weekly	33	27.98	33.85
Ozone	GRU	daily	32	4.78	5.87
Ozone	GRU	weekly	32	4.68	5.49
Ozone	LSTM	daily	32	5.02	5.95
Ozone	LSTM	weekly	32	4.90	5.85
PM10	GRU	daily	33	75.73	97.34
PM10	GRU	weekly	33	75.07	95.51
PM10	LSTM	daily	33	70.55	91.00
PM10	LSTM	weekly	33	62.80	78.35
PM2.5	GRU	daily	33	30.25	45.91
PM2.5	GRU	weekly	33	25.54	40.13
PM2.5	LSTM	daily	33	24.12	41.43
PM2.5	LSTM	weekly	33	18.63	33.00
SO2	GRU	daily	33	5.30	6.46
SO2	GRU	weekly	33	4.96	6.04
SO2	LSTM	daily	33	6.42	7.95
SO2	LSTM	weekly	33	7.27	8.86
Toluene	GRU	daily	32	10.88	11.87
Toluene	GRU	weekly	32	9.22	10.91
Toluene	LSTM	daily	32	11.31	12.03
Toluene	LSTM	weekly	32	10.98	12.15

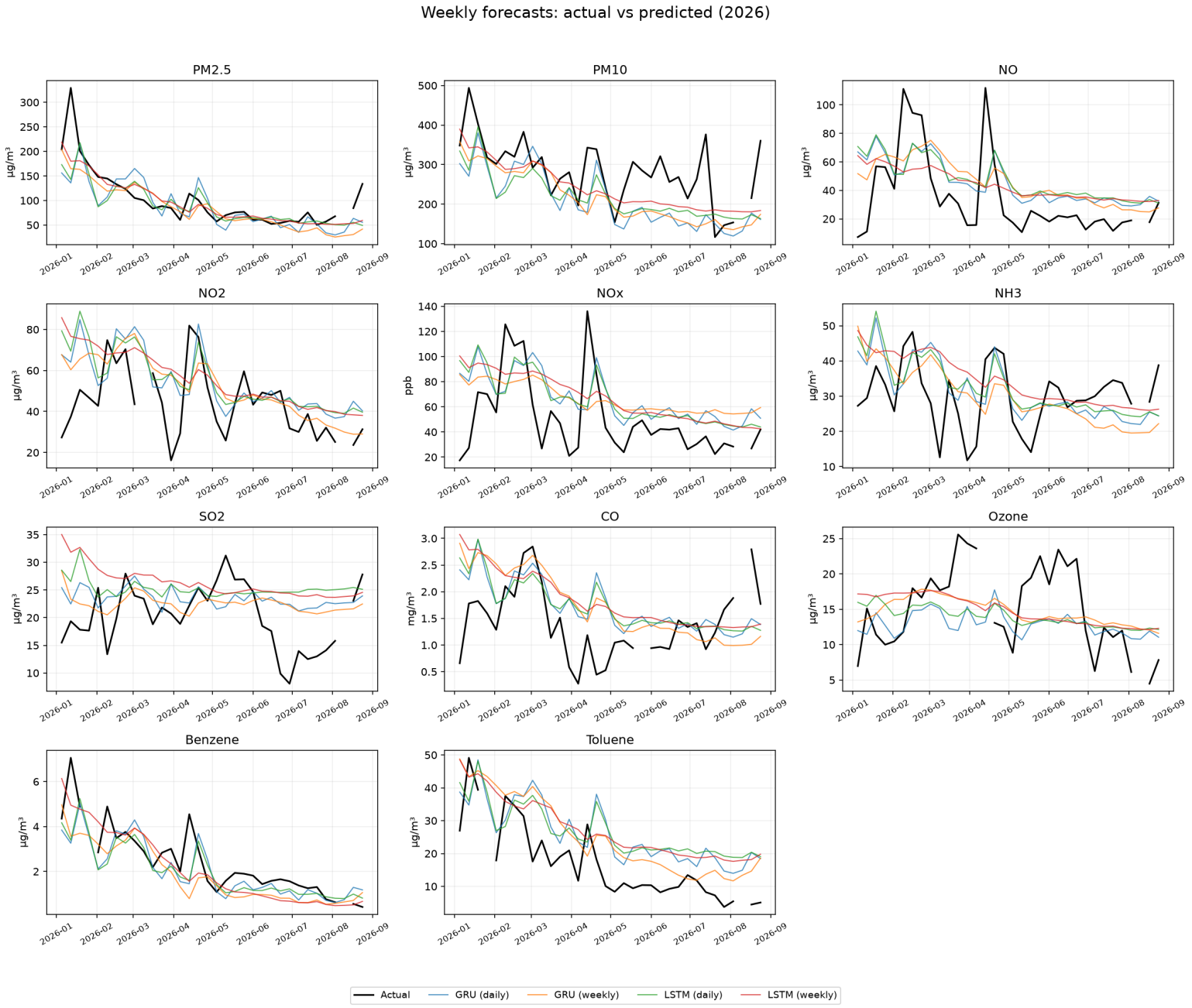
Weekly MAE comparison



Weekly RMSE comparison



Weekly actual and predicted values

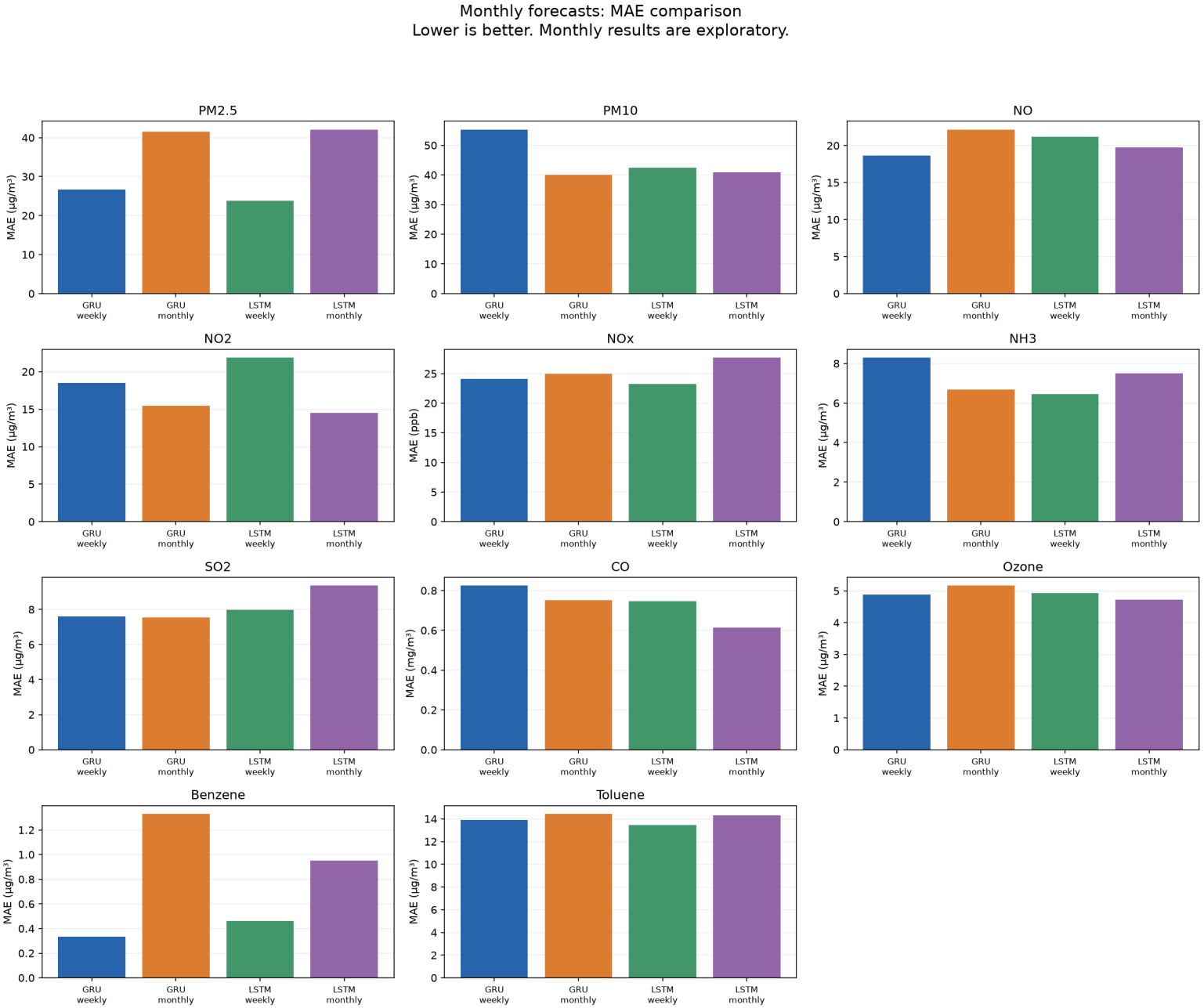


LSTM and GRU comparison monthly errors

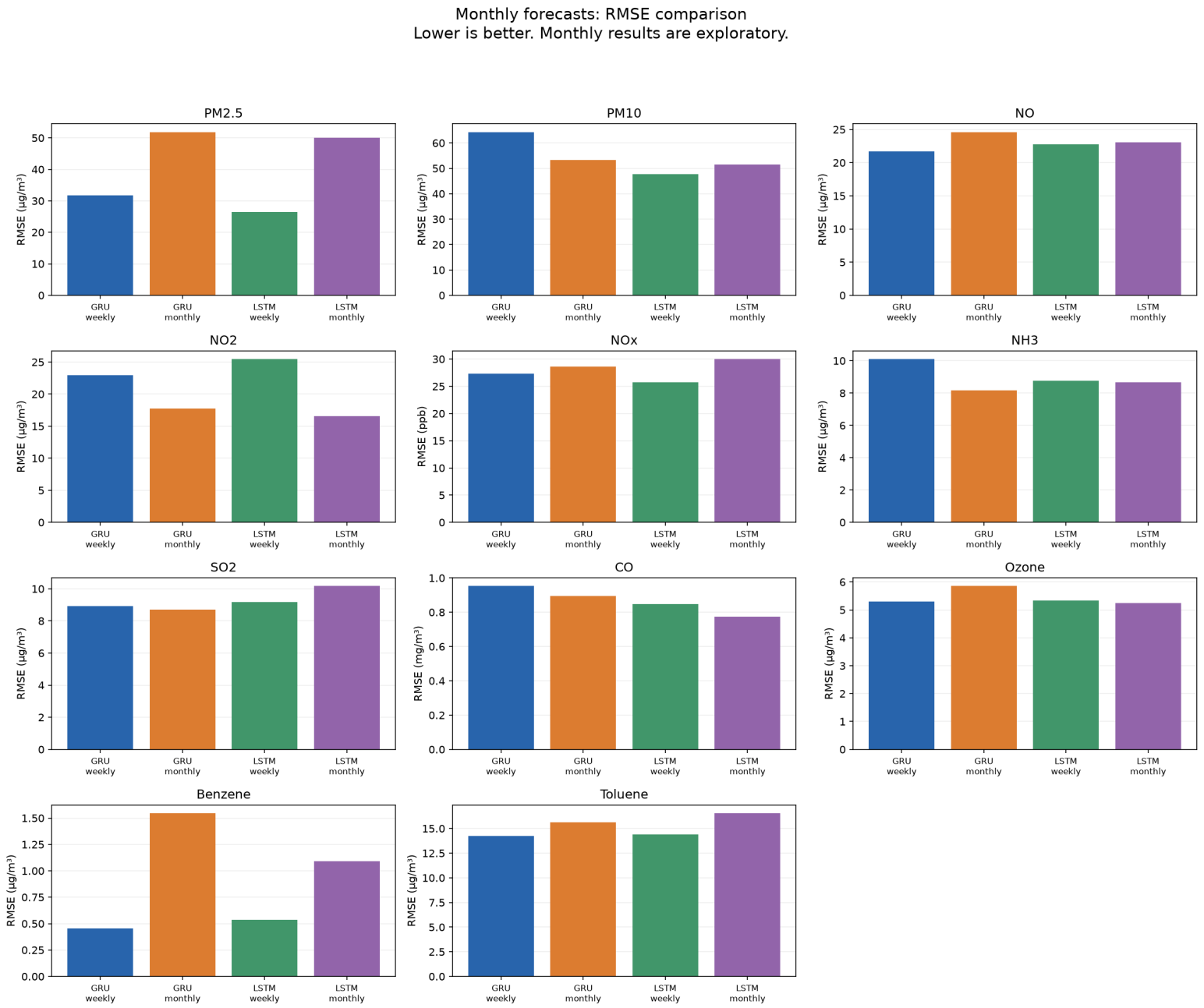
Original concentration units | N is the number of scored periods | Lower is better

Pollutant	Model	Input scale	N	MAE	RMSE
Benzene	GRU	monthly	8	1.33	1.55
Benzene	GRU	weekly	8	0.33	0.46
Benzene	LSTM	monthly	8	0.95	1.09
Benzene	LSTM	weekly	8	0.46	0.53
CO	GRU	monthly	8	0.75	0.89
CO	GRU	weekly	8	0.82	0.95
CO	LSTM	monthly	8	0.61	0.77
CO	LSTM	weekly	8	0.75	0.85
NH3	GRU	monthly	8	6.68	8.17
NH3	GRU	weekly	8	8.30	10.11
NH3	LSTM	monthly	8	7.50	8.68
NH3	LSTM	weekly	8	6.45	8.76
NO	GRU	monthly	8	22.17	24.62
NO	GRU	weekly	8	18.67	21.71
NO	LSTM	monthly	8	19.74	23.08
NO	LSTM	weekly	8	21.21	22.78
NO2	GRU	monthly	8	15.44	17.76
NO2	GRU	weekly	8	18.51	22.95
NO2	LSTM	monthly	8	14.49	16.56
NO2	LSTM	weekly	8	21.88	25.47
NOx	GRU	monthly	8	24.99	28.66
NOx	GRU	weekly	8	24.15	27.28
NOx	LSTM	monthly	8	27.71	30.01
NOx	LSTM	weekly	8	23.24	25.75
Ozone	GRU	monthly	8	5.17	5.85
Ozone	GRU	weekly	8	4.88	5.30
Ozone	LSTM	monthly	8	4.72	5.25
Ozone	LSTM	weekly	8	4.93	5.33
PM10	GRU	monthly	8	39.97	53.33
PM10	GRU	weekly	8	55.32	64.31
PM10	LSTM	monthly	8	40.88	51.59
PM10	LSTM	weekly	8	42.49	47.63
PM2.5	GRU	monthly	8	41.55	51.88
PM2.5	GRU	weekly	8	26.64	31.71
PM2.5	LSTM	monthly	8	42.10	49.99
PM2.5	LSTM	weekly	8	23.84	26.50
SO2	GRU	monthly	8	7.54	8.70
SO2	GRU	weekly	8	7.60	8.91
SO2	LSTM	monthly	8	9.36	10.17
SO2	LSTM	weekly	8	7.98	9.18
Toluene	GRU	monthly	8	14.45	15.62
Toluene	GRU	weekly	8	13.89	14.22
Toluene	LSTM	monthly	8	14.31	16.56
Toluene	LSTM	weekly	8	13.47	14.38

Monthly MAE comparison



Monthly RMSE comparison



Monthly actual and predicted values

